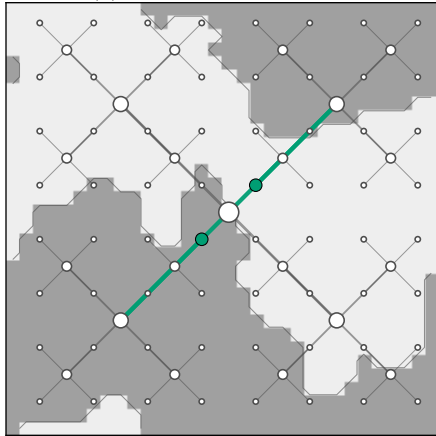
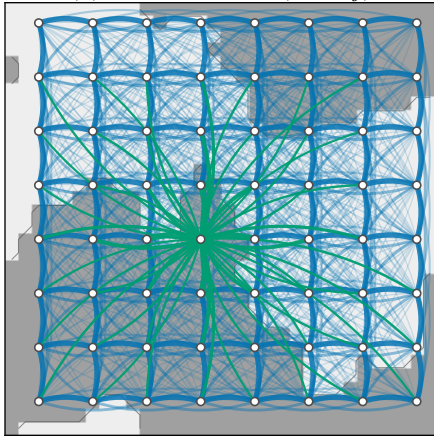


(a) tree: U-Net hierarchy



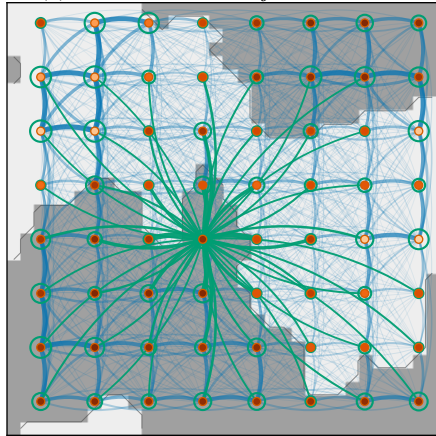
neighbouring cells across a block
boundary meet only at the root

(b) global, fixed: $\kappa_{\theta}(\mathbf{r}_i - \mathbf{r}_j)$



one input-independent pattern
around every cell (Fourier layer)

(c) rhizome: $\mathbf{A}_i \kappa_{\theta} \mathbf{B}_j$, update $t=4$



every cell sends ($\mathbf{B}$, fill) and listens
($\mathbf{A}$, ring) with its own weights